

# **Diphthongs in Dutch**

## *Phonological and phonetic perspectives on Standard Dutch and its dialects*

### **Abstract**

The core of the Dutch diphthong system are three diphthongs, one at each of the major places that Dutch also reserves for monophthongs: front unrounded /ɛi/, front rounded /ɥy/ and back rounded /au/. In this chapter we first discuss these three core diphthongs, their historical origin and their place in the current system.

We also compare these to two other sets. The first are the so-called ‘false diphthongs’ /uj, aj, oj, iw, ew/. These differ from the ‘real’ diphthongs in that they consist of vowel parts that are at *different* places of articulation (whereas real diphthongs are all in one place, differing only in height), and in their syllabification properties (true diphthongs allow non-coronal consonants in the coda, but false diphthongs do not).

The second set to which true diphthongs need to be compared are the mid vowels which in (Netherlands) Dutch tend to be diphthongized at least phonetically: /ej, øj/øw and ow/. Again the syllabification properties are different and in spite of their phonetics they phonologically still seem to fit completely in the set of monophthongs.

### **8.1 Introduction**

Dutch, a West Germanic language spoken by approximately 24 million people in the Netherlands, Belgium (Flanders), and Suriname, has a rich vowel system, even in comparison to other Germanic languages. The language features a complex set of diphthongs whose phonological status, phonetic realisation, and dialectal variation have been debated extensively in the scholarly literature. This chapter provides an overview of diphthongs in Dutch, covering both the standard language – in its Northern (Netherlandic) and Southern (Belgian) varieties – and the major dialect groups. The discussion addresses the phonological

representation of diphthongs, their phonetic properties, their historical development from earlier stages of the language, and their interaction with neighbouring segments.

The Dutch vowel system is notable for several reasons. First, it maintains a contrast between tense (long) and lax (short) monophthongs that interacts in complex ways with syllable structure. Second, it includes a set of phonemic diphthongs whose classification as ‘true’ or ‘false’ depends on one’s theoretical framework. Third, the mid tense vowels /eː, øː, oː/ are undergoing or have undergone diphthongisation in various regional and social varieties, a process that has attracted considerable sociolinguistic attention. Finally, the interaction of vowels with consonantal environment, particularly with /r/ and /l/, produces additional allophonic monophthongisation that blurs the boundary between monophthongs and diphthongs.

## 8.2 The Dutch Vowel System: An Overview

Before discussing diphthongs in detail, a brief sketch of the full vowel inventory of Standard Dutch is in order. Following Van Oostendorp (2000), Gussenhoven (1992, 2009) and Booij (1995), among others, Dutch has 13 monophthongal vowel phonemes and three diphthong phonemes. The monophthongs include the lax vowels /ɪ, ʏ, ɛ, ɔ, ʌ/ and the tense vowels /i, y, u, eː, øː, oː, aː/, along with the schwa vowel /ə/. The three phonemic diphthongs are /ɛi, œy, au/. Table 8.1 presents a schematic overview.

*Table 8.1. The Standard Dutch vowel system (after Gussenhoven, 2009)*

|                  | Front unr. | Front rnd. | Central | Back rnd. | Diphthongs |
|------------------|------------|------------|---------|-----------|------------|
| <b>Close</b>     | ɪ, ʏ       | y, ʏ       |         | u         |            |
| <b>Close-mid</b> | eː         | øː         |         | oː        |            |
| <b>Open-mid</b>  | ɛ          |            | ə       | ɔ         | ɛi, œy, au |
| <b>Open</b>      |            |            | aː      | ɑ         |            |

A central distributional fact about Dutch vowels is the so-called *checked–free* distinction: lax vowels (ɪ, ʏ, ɛ, ɔ, ʌ) occur only in closed syllables (they are ‘checked’), whereas tense vowels (i, y, u, eː, øː, oː, aː), the schwa vowel, and diphthongs may occur in both open and closed syllables (they are ‘free’). This distributional asymmetry has been analysed in terms of mora structure (Booij, 1995), skeletal slots (Kager, 1989), and licensing constraints (Van

Oostendorp 2000). The phonemic diphthongs pattern with the tense monophthongs as bimoraic nuclei, a fact that is important for understanding their phonological behaviour.

### **8.3 Transcription Conventions and Controversies**

The transcription of Dutch diphthongs is far from uniform in the literature, and the choices made reflect different theoretical commitments. The three phonemic diphthongs of Standard Dutch have been transcribed in at least the following ways: /ɛi, œy, au/ (Gussenhoven, 1992, 2009; Koopmans-van Beinum, 1980), /ɛi, œy, ɔu/ (Moulton, 1962; Collins & Mees, 2003), and /ɛi, œy, ou/ (various older sources). These differences are not merely notational. The choice between a tense second element (/i, y, u/) and a lax one (/ɪ, ʏ, ʊ/) reflects a claim about the phonetic endpoint of the glide. Gussenhoven (1992) argues that the second element is indeed a full close vowel, citing spectrographic evidence that the formant trajectories of the diphthongs reach the periphery of the vowel space. Collins and Mees (2003), on the other hand, use lax symbols on the grounds that the endpoint is typically not as close or as peripheral as the corresponding monophthong. In this chapter, we follow Gussenhoven's conventions, which have become the *de facto* standard in Dutch phonology and phonetics. The first element is equally contested. For the back diphthong, the choice between /ɑ/ and /ɔ/ as the first element again depends on how open one judges the onset to be. Acoustic studies (Adank et al., 2004; Jacobi, 2009) have shown considerable regional and individual variation in the starting points, which makes any single transcription inevitably an idealisation.

### **8.4 The Phonemic ('Real') Diphthongs: /ɛi, œy, au/**

#### **8.4.1 Phonological Status**

The three diphthongs /ɛi, œy, au/ are uncontroversially phonemic in Standard Dutch, as minimal pairs demonstrate: *bijt* /beit/ 'bites', *buut* /by:t/ 'lout', *buit* /bœyt/ 'boot', *bout* /baut/ 'bolt'. Phonologically, these diphthongs behave as single complex nuclei occupying two moras. Evidence for their monophonemic status includes the fact that they fill the same distributional slots as tense monophthongs: they occur in open syllables (*zij* /zei/ 'she'), before certain consonant clusters (*bijlst* is possible in principle), and in both stressed and unstressed positions.

Gussenhoven (2009) analyses the Dutch diphthongs as sequences of a vowel and a glide /j/ or /w/ underlyingly, but notes that this analysis faces challenges from the fact that the first

element of each diphthong is distinct from any monophthong. An alternative analysis treats them as specified for two sets of vocalic features, with the glide being an automatic consequence of the transition (Booij, 1995). Van Oostendorp (2000) proposes an account where the diphthongs consist of two skeletal positions dominated by a single nucleus, each associated with different melodic content.

#### **8.4.2 *Phonetic Realisation***

Acoustically, the Dutch diphthongs show the following general properties. The diphthong /*ei*/ begins in the open-mid front unrounded region and moves toward close front position.

Peeters (1991) and Adank et al. (2004) report that the onset F1 is typically around 600–700 Hz and F2 around 1700–1900 Hz for male speakers, with the offset approaching F1 values of 300–350 Hz and F2 values above 2000 Hz. The diphthong /*œy*/ is the rounded counterpart of /*ei*/, beginning in the open-mid front rounded area and moving toward close front rounded position; its F2 trajectory is somewhat lower due to lip rounding. The back diphthong /*au*/ starts in the open back region and moves toward a close back rounded target, showing a falling F1 trajectory and relatively stable low F2.

A notable phonetic feature of the Dutch diphthongs is the relatively long duration of the first element compared to the second. Van Heuven (1986) demonstrated that the steady-state portion of the first element accounts for roughly two-thirds of the total diphthong duration, with the glide transition occupying the final third. This asymmetry has been taken as evidence for a ‘falling’ prominence pattern, consistent with the phonological analysis of the first element as the syllable peak.

#### **8.4.3 *The Symmetry of the System***

The three diphthongs form a symmetrical set: /*ei*/ is front unrounded, /*œy*/ is front rounded, and /*au*/ is back rounded. This three-way pattern is typologically unusual; many languages with diphthongs lack a front rounded member, and front rounded vowels have been lost in many related languages (e.g., most English dialects). The symmetry is arguably maintained by the overall robustness of the front rounded vocalic series /*y*, *ɥ*, *ø*:/ in Dutch (Booij, 1995).

### **8.5 The Marginal (‘False’) Diphthongs**

In addition to the three phonemic diphthongs, Dutch has a number of vowel-glide sequences that are sometimes called ‘false’ or ‘marginal’ diphthongs. These include /ɑj/ in loanwords such as *ai* [ɑj] ‘exclamation’, /ɔj/ in *hoi* [hɔj] ‘hello’, /o:j/ in *mooi* [mo:j] ‘beautiful’, /uj/ in *foei* [fuj] ‘shame’, /iw/ in *nieuw* [niw] ‘new’ and /e:w/ in *leeuw* [le:w] ‘lion’. In all of these cases a back vowel is followed by a front /j/ and a front vowel by a back /w/. Their phonological status is fundamentally different from that of the true diphthongs.

The key argument for treating these as sequences of vowel + consonant rather than as diphthongs proper is distributional. Unlike the true diphthongs, which freely occur in word-final open syllables, the false diphthongs involve a glide consonant /j/ or /w/ that can be shown to occupy the syllable coda. Evidence comes from the fact that /j/ and /w/ in these sequences trigger resyllabification before vowel-initial suffixes: *mooie* [mo:.jə] ‘beautiful (inflected)’ shows that /j/ resyllabifies as an onset, which would not be expected if it were part of the nucleus (Booij, 1995). Furthermore, the vowel preceding /j/ in these sequences is typically a monophthong that occurs independently: /ɔ/ in *hoi*, /o:/ in *mooi*, /a:/ in *laai* ‘blaze’. The sequences with /w/, as in *nieuw* and *leeuw*, are similarly analysable as vowel + coda /w/ (Gussenhoven, 2009). In some dialects, the /w/ element has been lost entirely, yielding simple monophthongs.

Phonetically, false diphthongs differ from true diphthongs in their articulatory dynamics. Acoustic studies (Collins & Mees, 2003; Jacobi, 2009) have shown that the formant transitions in false diphthongs are steeper –they show a higher rate of spectral change per unit time – than those in true diphthongs. The true diphthongs /ɛi, œy, au/ exhibit a more gradual glide with a relatively long steady-state first element, whereas the false diphthongs involve a rapid transition to the glide target. This difference in temporal dynamics provides an additional phonetic criterion, beyond distributional evidence, for distinguishing the two categories.

The status of the false diphthongs is also relevant from a typological perspective. Cross-linguistically, the distinction between ‘true’ diphthongs (tautosyllabic vocalic sequences) and ‘false’ diphthongs (vowel + glide consonant sequences) is a recurring analytical challenge. Dutch provides a particularly clear case because the distributional evidence is robust: the true diphthongs pattern with tense vowels in the nucleus, while the false diphthongs pattern with VC sequences. The Dutch situation thus offers a benchmark for similar debates in other languages.

## **8.6 Diphthongisation of the Mid Tense Vowels**

### **8.6.1 The Phenomenon**

One of the most dynamic areas of change in contemporary Dutch phonology is the diphthongisation of the mid tense vowels /eː, øː, oː/. In conservative Standard Dutch, these vowels are relatively monophthongal, realised as [eː, øː, oː] with only slight diphthongal offglides. However, in many varieties of Netherlands Dutch (as opposed to Belgian Dutch or Suriname Dutch), they have developed into clearly diphthongal realisations: [ei], [øy], [ou]. At the same time, the ‘true’ diphthongs would be pronounced more open, and in particular /ei/ as [ɑi] or [ai]. This process has been described as ‘Polder Dutch’ (Stroop, 1998).

### **8.6.2 Sociolinguistic Dimensions**

Stroop (1998) was the first to draw widespread attention to this development, arguing that it originated among young, well-educated women in the Randstad area and constituted a new prestige norm. His term *Poldernederlands* captured the popular imagination but also proved controversial. Stroop characterised these speakers as a sociolinguistic avant-garde: opinion leaders, public intellectuals, authors, and media personalities whose cultural capital lent the innovative pronunciation an aura of modernity and authenticity. He argued that the innovation represented a change from below, operating beneath the threshold of conscious awareness, and that it constituted an implicit rejection of the traditional ABN (*Algemeen Beschaafd Nederlands*) norm, which was increasingly perceived as artificial and socially rigid. The physical opening of the vowels, in Stroop’s view, was a linguistic correlate of broader social emancipation.

Jacobi (2009) provided more systematic acoustic evidence, showing that the diphthongisation was indeed spreading across age groups and social classes in the Netherlands, with women consistently leading the change. Crucially, the phenomenon is largely absent from Belgian Standard Dutch, where the mid vowels remain monophthongal (Verhoeven, 2005). This North–South divergence is one of the most salient phonetic differences between the two national varieties and continues to widen as the Northern innovation progresses. Belgian Dutch seem to actively resist the ‘Hollandic’ diphthongisation, maintaining a more conservative vowel profile.

### **8.6.3 Phonological Implications**

The diphthongisation of the mid vowels creates a chain-shift scenario. As /e:/ moves towards [ei], it encroaches on the phonetic space of /ɛi/, which in turn lowers its starting point. Indeed, Stroop (1998) and Jacobi (2009) documented a concomitant lowering of the true diphthongs: /ɛi/ → [ai] or [ai], /œy/ → [ʌy] or [əy], and /au/ → [ʌu]. This constitutes a classic push chain, comparable to the English Great Vowel Shift in its structural logic, though far more limited in scope (Gussenhoven, 2009). Not all scholars agree that the lowering of the true diphthongs is causally driven by the diphthongisation of the mid vowels; some argue for independent but parallel developments (Van de Velde & van Hout, 2003).

An important phonological point is that the lowering of the onset of the true diphthongs is a purely qualitative change: the duration of the lowered diphthong remains stable and does not encroach on the domain of the ‘false’ diphthongs such as /a:j/ in *haai* ‘shark’. The innovative [ai] for /ɛi/ retains the temporal structure of a phonemic diphthong and is not confused with a long monophthong + glide consonant sequence. This durational stability suggests that the change targets spectral quality rather than phonological structure (Jacobi, 2009).

## **8.7 Historical Development**

The diphthongs of modern Dutch have diverse historical origins. The present-day phonemic diphthongs /ɛi, œy, au/ arose during the Middle Dutch period (roughly 1150–1500) through the diphthongisation of the Old Dutch long high vowels /i:, y:, u:/. This development parallels similar changes in other West Germanic languages, including the High German diphthongisation (/i:/ > /ai/, etc.) and the English Great Vowel Shift, though the Dutch changes are generally dated earlier than the English ones (Van Bree, 1987; Van Loey, 1970). The Old Dutch diphthongisation proceeded in stages. The first step, occurring in the 12th–13th centuries in the Hollandic and Brabantian dialects, affected /i:/, which lowered and broke to [ei] and subsequently to [ɛi]. The rounding counterpart /y:/ underwent a parallel change to [œy], while /u:/ became [au] (Van Reenen & Wattel, 1992). The chronology of these changes is debated: some scholars argue for a simultaneous shift of all three vowels (a chain shift), while others posit sequential changes with /i:/ leading.

A second major historical source of diphthongs is l-vocalisation. In the transition from Middle to Modern Dutch, /l/ in coda position before dental consonants was vocalised in the Hollandic dialects: older sequences /-ald, -old, -olt/ yielded /-aud, -aut/, adding to the functional load of /au/. Thus Modern Dutch *goud* ‘gold’ (< Middle Dutch *golt*), *oud* ‘old’ (< *old/ald*), and *zout* ‘salt’ (< *solt*) all acquired /au/ through this process. The orthographic

distinction between *ou* and *au* in Modern Dutch (e.g., *koud* ‘cold’ vs. *blauw* ‘blue’) preserves a historical difference that had been phonetically neutralised by the 19th century at the latest: both spellings now represent the single phoneme /au/. A parallel merger affected the front diphthong: the spellings *ei* and *ij*, originally representing distinct sounds (/ei/ from Germanic \*ai and /ei/ from diphthongised /i:/), collapsed into a single phoneme /ei/ in Standard Dutch, though some dialects (e.g., parts of Brabant) maintain a vestigial distinction (Van Loey, 1970).

Importantly, the diphthongisation did not affect all dialect areas equally. The peripheral dialects (West Flemish, Limburgish, and the northeastern Low Saxon varieties) largely preserved the old monophthongs or developed differently. In West Flemish, for instance, Old Dutch /i:/ was preserved as a long monophthong or raised rather than diphthongised, yielding modern [ɛ:] or [i:] in words where Standard Dutch has /ei/ (Taeldeman, 1979; De Schutter, 1994). These dialect differences are discussed further in Section 8.10.

The contemporary diphthongisation of the mid vowels discussed in Section 8.6 can be viewed from this historical vantage point as the resumption of the medieval chain shift after a pause of several centuries. The 14th-century Brabantian Expansion drove the initial diphthongisation of /i:, y:, u:/, which subsequently lowered through the 15th and 16th centuries until the onset of the diphthongs stabilised in the mid-low region [ɛ, œ, ɑ] during the 17th century. This 17th-century phonetic steady state became codified as the standard pronunciation. Stroop (1998) argues that Polder Dutch represents the ‘unfreezing’ of this historical pause: the continued lowering of the diphthong onsets, and the concomitant diphthongisation of the mid vowels that fill the vacated phonetic space, are the natural continuation of a process set in motion some 600 years ago.

## **8.8 Interaction with Neighbouring Segments**

### **8.8.1 Diphthongs and /r/**

The interaction between diphthongs and /r/ is one of the most striking areas of Dutch phonology and phonetics. The most fundamental observation is a co-occurrence restriction: the phonemic diphthongs /ei, œy, au/ do not occur before tautosyllabic /r/ in native Dutch vocabulary. Words such as *bier* ‘beer’, *deur* ‘door’, *buur* ‘neighbour’, and *uur* ‘hour’ have the long monophthongs /i:, ø:, y:, u:/ rather than diphthongs. This is not an accident of the lexicon but a reflection of the historical process itself: the medieval diphthongisation of /i:, y:,

u:/ that produced /ei, œy, au/ was blocked before /r/. The result is a synchronic phonotactic gap: diphthongs and tautosyllabic /r/ are incompatible. Booij (1995) and Gussenhoven (2009) note this restriction, which sets Dutch apart from languages like English, where diphthong + /r/ sequences are common (*fire, hour*) (although obviously also in English a schwa typically intervenes between the diphthong and r).

Dutch /r/ is itself highly variable, with realisations ranging from alveolar trills [r] and taps [ɾ] to uvular trills [ʀ] and fricatives [ʁ], as well as approximants [ɹ] and even vocalised variants [ə] (Van de Velde & van Hout, 2003; Sebregts, 2014). The vocalisation of /r/ after vowels, increasingly common in Randstad Dutch (Sebregts, 2014), creates additional complications. When coda /r/ is realised as a schwa-like vowel [ə], the preceding long monophthong may acquire a centering offglide, yielding sequences like [iə] for /i:r/ in *bier* or [øə] for /ø:r/ in *deur*. These surface diphthongs are phonetically real but phonologically distinct from the true diphthongs: they arise from r-vocalisation rather than from a diphthongal nucleus, and /r/ will still surface in certain contexts (like plural, which is always [d ø:rə]).

### **8.8.2 Diphthongs and /l/**

The interaction with /l/ is equally consequential. In Randstad Dutch, /l/ in coda position is increasingly vocalised to [ʊ] or [u], creating new diphthongal sequences on the surface: *volk* [vʊok] ‘people’, *melk* [mɛʊk] ‘milk’. When the phonemic diphthongs precede /l/, complex triphthongal realisations may result, as in *vuil* /vœyl/ → [vœyʊ] (Collins & Mees, 2003). The vocalisation of /l/ is less advanced than that of /r/ and shows considerable social and regional stratification.

A particularly interesting finding, documented by Voeten (2021), is that coda /l/ blocks the ongoing diphthongisation of the mid tense vowels. In varieties where /e:, ø:, o:/ are diphthongised to [ei, øy, ou] in most environments, they remain monophthongal before /l/: *veel* ‘much’ retains [e:] even in speakers who produce [ei] in *been* ‘leg’. This blocking effect may be related to the articulatory conflict between the upward glide movement of the diphthong and the tongue-body retraction required for dark or vocalised /l/. The result is a synchronic conditioning environment that constrains the spread of the diphthongisation process.

### **8.8.3 Effects of Following Nasals and Obstruents**

Before voiceless obstruents, the diphthongs tend to be shorter and may show less extreme formant movement. Slis and Cohen (1969) documented the durational effects of voicing in the following consonant on Dutch vowels, including diphthongs. Before nasals, diphthongs may show nasalisation of the first element, and the glide transition may be compressed. These coarticulatory effects, while phonetically significant, have not been shown to be phonologised in Standard Dutch, though dialectal developments in some varieties suggest incipient phonologisation (De Wulf & Taeldeman, 2001).

### **8.9 Phonotactic Distribution**

The phonemic diphthongs /*ei*, *œy*, *au*/ share the distribution of the tense monophthongs, appearing freely in open and closed syllables alike. They occur word-initially (*ui* /*œy*/ ‘onion’), word-medially, and word-finally. In codas, they may be followed by up to two consonants in native vocabulary, provided the last consonants are coronal: *bijlſt* would be possible, though such heavy codas are rare, but something like \*[*beimp*] is not, even though [*mp*] after lax vowels (*ramp* [*ramp*] ‘disaster’). Booij (1995) formulates the constraint that bimoraic nuclei (tense vowels and diphthongs) may be followed by maximally two coda consonants, whereas monomoraic (lax) nuclei may be followed by up to three.

In onset position, diphthongs interact with the general constraints on Dutch syllable structure. Dutch permits complex onsets of up to three consonants (e.g., /*str*/), and these may precede diphthongs without restriction: *strijd* /*streit*/ ‘battle’, *struik* /*stroeyk*/ ‘bush’. However, certain onset+diphthong combinations are lexically rare or absent, not because of phonotactic constraints per se but due to historical accident. The combination of velar nasals with diphthongs, for instance, does not arise because /*ŋ*/ does not occur in onset position in Dutch (like in English).

The distributional overlap between diphthongs and tense monophthongs extends to morphophonological alternations. In Dutch nominal and verbal morphology, there are no alternations between diphthongs and monophthongs of the kind found in, say, English (e.g., *divine* ~ *divinity*). The diphthongs are stable across inflectional and derivational paradigms, which reinforces their status as single phonological units rather than derived sequences. This stability contrasts with the allophonic diphthongisation of mid vowels, which is a gradient phonetic phenomenon without morphophonological consequences.

### **8.10 Dialectal Variation**

*Table 8.2. Approximate realisations of the diphthongs across Dutch dialect regions*

| <b>Dialect region</b> | <b>/ɛi/</b>  | <b>/œy/</b> | <b>/au/</b> | <b>Mid vowels</b>  |
|-----------------------|--------------|-------------|-------------|--------------------|
| Std. Northern         | [ɛi]         | [œy]        | [au]        | [e:] ~ [ei]        |
| Polder Dutch          | [ai] ~ [ai̯] | [əy] ~ [ʌy] | [ʌu]        | [ei], [øy], [ou]   |
| Belgian Std.          | [ɛ:]         | [œ:]        | [ɔ:]        | [e:], [o:]         |
| West Flemish          | [ɛ:] ~ [i:]  | [œ:] ~ [y:] | [ɔ:] ~ [u:] | [e:], [o:]         |
| Limburgish            | [ɛi] ~ [i:]  | [y:]        | [au] ~ [ɔu] | [ei], [ou] (tonal) |
| Low Saxon             | [ɛi] ~ [ai̯] | [œy] ~ [y:] | [ɔu] ~ [ou] | [e:], [o:]         |

### **8.10.1 Belgian Dutch and Flemish Dialects**

The most salient North–South difference in the Dutch-speaking area concerns the realisation of the phonemic diphthongs, as noted above. In Belgian Standard Dutch, these are typically realised as long monophthongs [ɛ:, œ:, ɔ:] rather than as diphthongs proper (Verhoeven, 2005). This monophthongal realisation reflects the Brabantian dialectal substrate. In the traditional West Flemish dialects, the correspondences to Standard Dutch diphthongs are even more divergent: Old Dutch /i:/ was either preserved or raised, and West Flemish may have [i:] or [ɛ:] where Standard Dutch has /ɛi/ (Taeldeman, 1979). East Flemish and Brabantian patterns show intermediate realisations.

### **8.10.2 Limburgish**

Limburgish, spoken in the southeastern Netherlands and northeastern Belgium, occupies a special position because it is a tonal dialect group. The lexical tone distinction (‘sleeptoon’ vs. ‘stoottoon’, or Accent 1 vs. Accent 2) interacts with vowel quality and diphthongisation. Gussenhoven and Aarts (1999) show that in Maastrichtian, for instance, the tonal accent affects the trajectory of diphthongs: Accent 1 diphthongs may show a more gradual glide, while Accent 2 diphthongs may have a more abrupt transition. Limburgish also preserves some old monophthongs where Standard Dutch has diphthongs, particularly /y:/ for Standard Dutch /œy/ in certain lexical items. Additionally, Limburgish has developed diphthongs of its own that are absent in the standard language, contributing to its remarkably rich vowel inventory (Gussenhoven & Aarts, 1999).

### **8.10.3 Low Saxon Dialects**

The northeastern Low Saxon dialects (Gronings, Twents, Drents, etc.) present yet another pattern. Many of these dialects did not undergo the medieval diphthongisation or did so only partially. Kloeke (1927) mapped the isoglosses separating monophthongal and diphthongal realisations of the old long high vowels, showing a gradual transition from the innovative Hollandic centre to the conservative northeast. In modern Low Saxon dialects, one finds both preserved monophthongs ([iː, yː, uː]) and locally innovated diphthongs that differ from the standard realisations. The interaction with the standard language through contact and koineisation has created complex patterns of variation (Nerbonne & Heeringa, 2001).

### **8.10.4 Hollandic Dialects and Urban Vernaculars**

The Hollandic dialects, forming the historical basis of Standard Dutch, are the most innovative with respect to diphthongisation. In addition to the general diphthongisation of /eː, øː, oː/ discussed in Section 8.6, urban vernaculars in Amsterdam, Rotterdam, and The Hague show further developments. These include extreme lowering of the phonemic diphthongs (/ɛi/ → [aə], /au/ → [ɔə]) and centralisation patterns that have been linked to multicultural urban speech varieties (Grondelaers & Van Hout, 2011). In the Amsterdam vernacular, for example, the diphthong /ɛi/ may be realised as [aə] before /r/, creating considerable auditory distance from the standard realisation.

### **8.10.5 Zeelandic**

The Zeelandic dialects, spoken in the southwestern Netherlands, present an interesting transitional case between Hollandic and Flemish patterns. Zeelandic preserves some older monophthongal forms while also showing innovations not found in either the northern or southern standard varieties. De Wulf and Taeldeman (2001) document Zeelandic realisations of the diphthong phonemes that range from long monophthongs to wide diphthongs depending on the specific dialect and the phonetic context.

### **8.10.6 A Note on Afrikaans**

Afrikaans, which developed from 17th-century Dutch settler dialects, has transformed the Dutch diphthong system in revealing ways. The three Dutch diphthongs were largely monophthongised in Afrikaans: /ɛi/ > /ɛɪ/ (with a much reduced glide) or /ae/, /œy/ > /œɪ/,

and /au/ > /əu/ or /ɔu/ (Wissing, 2012). At the same time, Afrikaans developed new diphthongs from Dutch long monophthongs. These developments, while beyond the scope of this chapter, provide valuable evidence for reconstructing the phonetic values of 17th-century Dutch diphthongs; see Andries Coetzee & Martin Moessmer (this volume) for more information.

### **8.11 Cross-Linguistic Perspective**

Since this chapter appears in a volume on diphthongs across the world's languages, a brief comparison with neighbouring Germanic languages is in order. Standard German has three diphthongs /ai, ɔʏ, au/ that are roughly cognate with Dutch /ei, œy, au/. However, the German diphthongs have a markedly lower starting point: German /ai/ begins near open [a], whereas Dutch /ei/ begins in the open-mid region [ɛ] (though the Polder Dutch lowering is closing this gap). German /au/ likewise starts lower than Dutch /au/. The front rounded diphthong /ɔʏ/ in German is backed relative to Dutch /œy/. In several Limburgish dialects, the realisations of the Dutch diphthongs approach the German values, reflecting the geographic and linguistic continuum between the two languages.

English provides another instructive comparison. English /eɪ/ (as in *day*) is phonetically similar to Dutch /ei/, though the English diphthong starts slightly higher. English /aɪ/ (as in *price*) has no standard Dutch counterpart but matches the innovative Polder Dutch [ai]. English /aʊ/ (as in *mouth*) closely approximates Dutch /au/. Crucially, English lacks a front rounded diphthong entirely, making Dutch /œy/ typologically distinctive. West Frisian, a closely related language spoken in the northern Netherlands, has its own diphthong set (/ai, ɔi, uy/) that arose independently but shows convergence with Dutch diphthongs in bilingual communities (Peeters, 1991).

### **8.12 Conclusion**

The diphthong system of Dutch is characterised by a layered complexity that makes it a particularly instructive case in any cross-linguistic survey. The language maintains a three-way contrast among phonemic diphthongs /ei, œy, au/ that is both typologically unusual (especially in its inclusion of a front rounded member) and phonologically well-integrated into the vowel system. Alongside these 'true' diphthongs, the language features 'false' diphthongs that are better analysed as vowel-glide sequences, as well as an ongoing process

of diphthongisation of the mid vowels that is reshaping the phonetic landscape of Northern Dutch.

The dialectal picture adds further richness. The variation between Belgian monophthongal realisations and Northern diphthongal ones, the tonal interactions in Limburgish, the conservative patterns in Low Saxon, and the innovative chain shifts in Hollandic urban speech all demonstrate that diphthongs in Dutch are not a static inventory but a dynamic system subject to continuous change. The interaction with consonantal environment – particularly with /r/ and /l/ – further blurs the boundaries of the category. Any synchronic snapshot of Dutch diphthongs is therefore necessarily incomplete without attention to the diachronic and dialectological dimensions that have shaped the present-day system.

## References

- Adank, P., van Hout, R., & Smits, R. (2004). An acoustic description of the vowels of Northern and Southern Standard Dutch. *Journal of the Acoustical Society of America*, 116(3), 1729–1738. <https://doi.org/10.1121/1.1779271>
- Booij, G. (1995). *The phonology of Dutch*. Clarendon Press.
- Collins, B., & Mees, I. M. (2003). *The phonetics of English and Dutch* (5th ed.). Brill.
- De Schutter, G. (1994). Dutch. In E. König & J. van der Auwera (Eds.), *The Germanic languages* (pp. 439–477). Routledge.
- De Wulf, C., & Tældeman, J. (2001). Apocope en deletie van de sjwa in de Vlaamse en Zeeuwse dialecten. *Taal en Tongval*, 53(1), 1–51.
- Grondelaers, S., & Van Hout, R. (2011). The standard language situation in the Low Countries: Top-down and bottom-up variations on a diaglossic theme. *Journal of Germanic Linguistics*, 23(3), 199–243.
- Gussenhoven, C. (1992). Dutch. *Journal of the International Phonetic Association*, 22(2), 45–47. <https://doi.org/10.1017/S002510030000459X>
- Gussenhoven, C. (2009). Vowel duration, syllable quantity and stress in Dutch. In K. Hanson & S. Inkelas (Eds.), *The nature of the word: Studies in honor of Paul Kiparsky* (pp. 181–198). MIT Press.
- Gussenhoven, C., & Aarts, F. (1999). The dialect of Maastricht. *Journal of the International Phonetic Association*, 29(2), 155–166.
- Jacobi, I. (2009). *On variation and change in diphthongs and long vowels of spoken Dutch* [Doctoral dissertation, University of Amsterdam].
- Kager, R. (1989). *A metrical theory of stress and destressing in English and Dutch*. Foris.
- Kloeke, G. G. (1927). *De Hollandsche expansie in de zestiende en zeventiende eeuw en haar weerspiegeling in de hedendaagsche Nederlandsche dialecten*. Martinus Nijhoff.
- Koopmans-van Beinum, F. J. (1980). *Vowel contrast reduction: An acoustic and perceptual study of Dutch vowels in various speech conditions* [Doctoral dissertation, University of Amsterdam].
- Moulton, W. G. (1962). The vowels of Dutch: Phonetic and distributional classes. *Lingua*, 11, 294–312.
- Nerbonne, J., & Heeringa, W. (2001). Computational comparison and classification of dialects. *Dialectologia et Geolinguistica*, 9, 69–83.

- Peeters, W. J. M. (1991). *Diphthong dynamics: A cross-linguistic perceptual analysis of temporal patterns in Dutch, English, and German* [Doctoral dissertation, Utrecht University].
- Sebregts, K. (2014). *The sociophonetics and phonology of Dutch r* [Doctoral dissertation, Utrecht University]. LOT.
- Slis, I. H., & Cohen, A. (1969). On the complex regulating the voiced–voiceless distinction. *Language and Speech*, 12(2), 80–102.
- Stroop, J. (1998). *Poldernederlands: Waardoor het ABN verdwijnt*. Bert Bakker.
- Taeldeman, J. (1979). *Het klankpatroon van de Vlaamse dialecten: Een synchrone beschrijving*. Rijksuniversiteit Gent.
- Van Bree, C. (1987). *Historische grammatica van het Nederlands*. Foris.
- Van de Velde, H., & van Hout, R. (2003). Pronunciation of Dutch. In J. K. Chambers, P. Trudgill, & N. Schilling-Estes (Eds.), *The handbook of language variation and change* (pp. 239–269). Blackwell.
- Van Heuven, V. J. (1986). Some acoustic characteristics of Dutch diphthongs. In F. Beukema & A. Hulk (Eds.), *Linguistics in the Netherlands 1986* (pp. 97–106). Foris.
- Van Loey, A. (1970). *Schönfelds historische grammatica van het Nederlands* (8th ed.). Thieme.
- Van Oostendorp, M. (2000). *Phonological projection: A theory of feature content and prosodic structure*. Mouton de Gruyter.
- Van Reenen, P., & Wattel, E. (1992). The phonetic basis of the Old Dutch monophthongization. In J. van Marle (Ed.), *Historical linguistics 1991* (pp. 303–320). John Benjamins.
- Verhoeven, J. (2005). Belgian Standard Dutch. *Journal of the International Phonetic Association*, 35(2), 243–247.
- Voeten, C. C. (2021). Regional variation in ongoing sound change: The case of the Dutch diphthongs. *Journal of Linguistic Geography*, 9(2), 162–177.
- Wissing, D. (2012). Afrikaans. *Journal of the International Phonetic Association*, 42(3), 354–358.